UNSUPERVISED LEARNING CASE STUDY

Customer Base Typology

A six-month behavioral segmentation of **20,000 customers**

2025-07-01 — 2025-12-31 / ANALYTICAL PRESENTATION

01 / PROBLEM

One tariff hides six materially different customer states

20,000

customers in the source base, currently served under one **standard** tariff.

5

active behavioral clusters created from spending, financial standing, and engagement.

1

operational inactive state for customers with no transaction in the six-month window.

Purpose. Turn an undifferentiated base into a decision system for targeting, offers, retention, and tariff architecture.

02 / WINDOW

The observation window is six months, not twenty-four

2025-07-01 — 2025-12-31

A shorter common window avoids describing a customer who never existed in the data.

3,424,008
transactions in the window

119,176
balance observations

119,176
monthly revenue observations

359
customers with no transaction in six months

03 / AXES

Three behavioral axes define the typology

AXIS 01

Spending behavior

Transaction count, total and average amounts, dispersion, channel & MCC-category shares.

AXIS 02

Financial standing

Mean, SD, min, max and coefficient of variation of monthly balance.

AXIS 03

Engagement

Number of distinct products per customer.

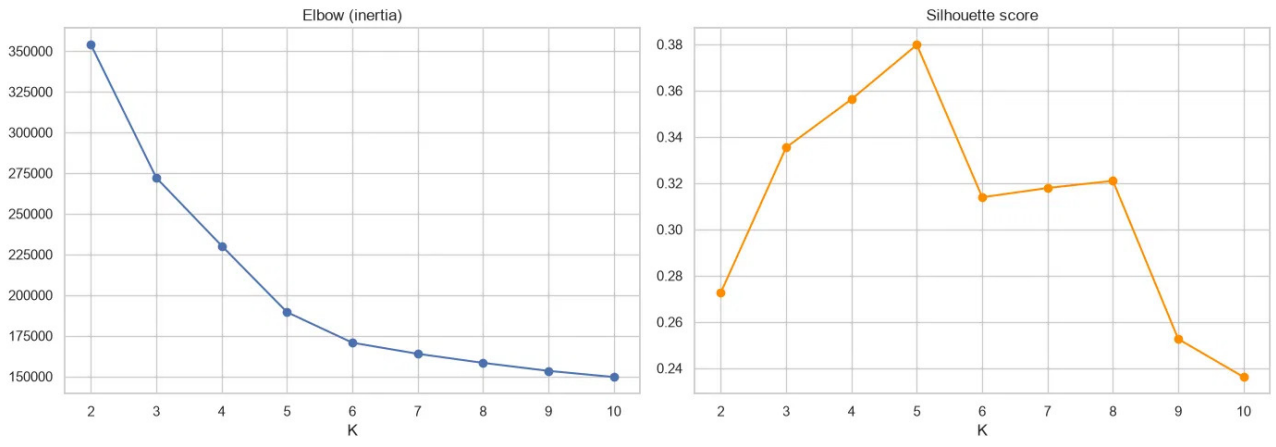
Excluded from clustering. `net_revenue_uzs` used only after clustering for valuation.

The pipeline protects the result from scale and inactivity

01	02	03	04
Aggregate	**Separate inactivity**	**Normalize**	**Evaluate**
Build customer-level transaction, balance, product, channel, and MCC features.	Hold out the **359** customers with no transaction during the six-month window.	Apply **log1p** to skewed monetary features and standardize the active matrix.	Fit K-Means with **random_state=42**, **n_init=10**, and K=2…10.

Control principle. Revenue is not an input feature; it is measured after clustering as an external business outcome.

Five clusters balance separation and interpretability



elbow and silhouette evaluation for K=2…10.

K	Inertia	Silhouette
2	353,857.87	0.27
3	272,086.27	0.34
4	230,312.70	0.36
5	189,495.01	0.38
6	170,732.87	0.31
7	163,901.68	0.32
8	158,323.84	0.32
9	153,416.72	0.25
10	149,625.24	0.24

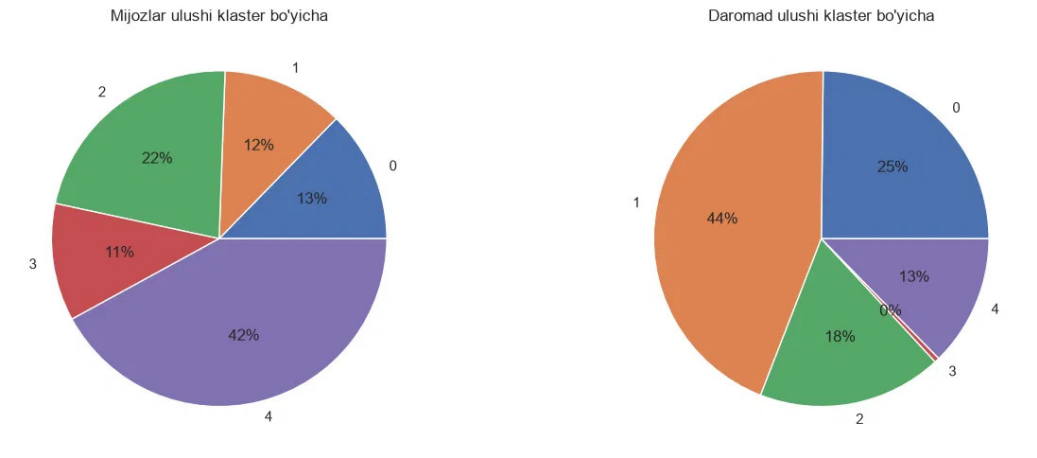
Decision. K=5 has the highest silhouette score; larger K values reduce separation and fragment the business story.

The five active segments have distinct behavioral signatures

SEGMENT	CUSTOMERS	DEFINING EVIDENCE	ADDITIONAL PROFILE VALUES
0 — Diversified high-balance users	2,493	POS share **0.56**; travel **0.16**; electronics **0.16**	`txn_cnt=3.99`; mean balance **16.69**; products **2.12**
1 — High-value mixed-channel users	2,306	`txn_cnt=6.16`; ATM **0.27**; e-commerce **0.27**	mean balance **15.35**; products **2.33**
2 — E-commerce and everyday spenders	4,359	`txn_cnt=5.61`; e-commerce **0.47**; restaurants **0.21**; transport **0.18**	mean balance **14.39**; products **1.69**
3 — Low-activity POS users	2,225	`txn_cnt=2.21`; POS **0.59**; pharmacy **0.10**	mean balance **12.67**; products **0.91**
4 — ATM-led mainstream users	8,258	ATM **0.37**; POS **0.52**; utilities **0.13**	mean balance **13.97**; products **1.17**

Segment labels are analytical summaries of the exact notebook profile values, not source-table fields.

Revenue concentration is not the same as customer concentration

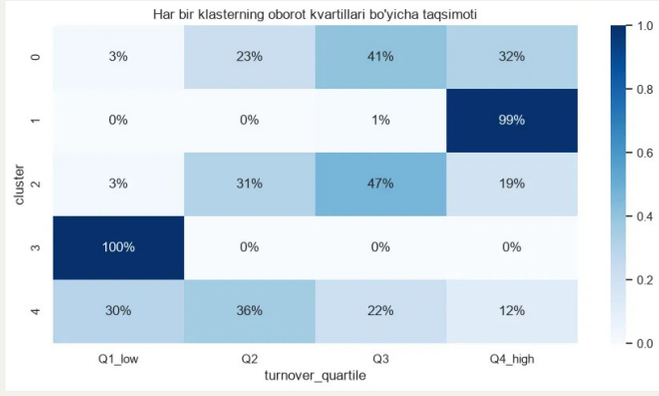


customer share and six-month revenue share by cluster.

CLUSTER	CUSTOMERS	CUST. SHARE	REVENUE (UZS)	REV. SHARE
0	2,493	12.70%	2,980,796,573	24.80%
1	2,306	11.70%	5,317,419,729	44.30%
2	4,359	22.20%	2,135,466,104	17.80%
3	2,225	11.30%	56,153,620	0.50%
4	8,258	42.00%	1,518,427,347	12.60%

Key readout. Cluster 1 is 11.70% of customers but 44.30% of six-month revenue; Cluster 4 is 42.00% of customers but 12.60% of revenue.

The typology adds behavior beyond turnover quartiles

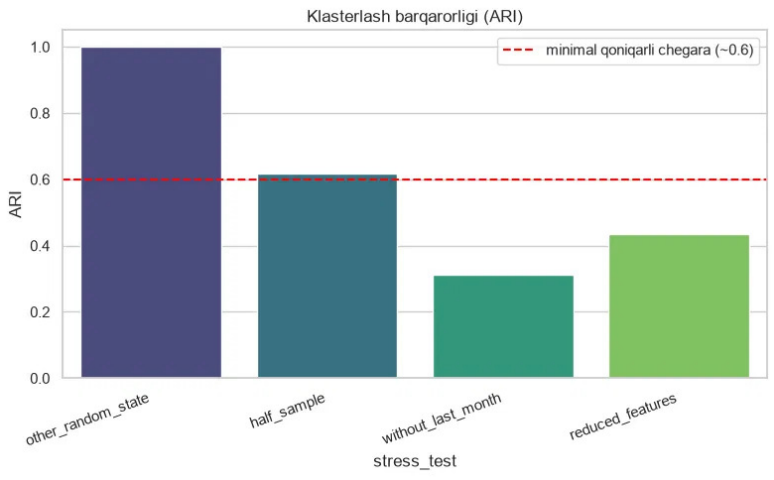


cluster distribution across turnover quartiles.

CLUSTER	Q1 LOW	Q2	Q3	Q4 HIGH
0	87	584	1,033	789
1	0	0	23	2,283
2	118	1,370	2,063	808
3	2,223	2	0	0
4	2,483	2,954	1,791	1,030

Interpretation. Cluster 3 is almost entirely Q1-low, while Cluster 4 spans all four quartiles; behavior is not reducible to turnover rank.

Stable to random initialization, weaker to time and feature changes



ARI stability under four stress tests.

STRESS TEST	ARI
Other `random_state`	**1.00**
Half sample	**0.62**
Without last month	**0.31**
Reduced feature set	**0.43**

Interpretation. The solution is identical under a different random state, but sensitive to the observation window and to channel/MCC features.

A new customer can be assigned with 98.2% holdout accuracy

98.2%

simulated success rate on customers treated as new at scoring time.

01 Activity gate
No transaction in six months → inactive state.

02 Feature window
Compute the same six-month customer features.

03 Classifier
Use Random Forest with `n_estimators=300` and `random_state=42`.

04 Validation
Holdout test: **4,911** customers; accuracy **0.982**.

Operational rule. Inactive customers bypass behavioral classification; active customers are assigned to one of the five clusters.

The segmentation changes the product playbook

ANALYTICAL STATE	RECOMMENDED DECISION
Diversified high-balance users	Premium relationship proposition; deepen travel, electronics, and multi-product usage.
High-value mixed-channel users	Protect and expand the highest-value base with premium service and cross-sell.
E-commerce and everyday spenders	Build digital commerce, restaurant, and transport propositions.
Low-activity POS users	Use activation, pharmacy, and everyday-spend nudges before premium pricing.
ATM-led mainstream users	Improve cash-to-digital migration and attach utilities or adjacent products.
Inactive state	Trigger reactivation journeys; do not force this state into behavioral clustering.

Replace one tariff for everyone with differentiated acquisition, engagement, retention, and pricing actions.